

Return values and local state

Project: Python readiness portfolio

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Functions and arguments

Without opening the old lesson, explain one key idea from Functions and arguments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Loops and trace tables

Without opening the old lesson, explain one key idea from Loops and trace tables. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Expressions and operator order

Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace separate calls for Return values and local state. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Return values and local state. Proposed explanation: “Printing a value returns that value”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Use returns and keep local state local

Trace both calls and the final expression. If return total is removed from the function, does this final print complete? Would a separate print(total) outside the function have access to its local variable?

```
def twice(n):  
    total = 2 * n  
    return total  
  
first = twice(3)  
second = twice(5)  
  
print(first, second, first + second)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Separate returning from displaying

Using the correct worked definition, execute result = remaining(15, 6), then doubled = result * 2. State both stored values and what the script displays before a print statement is added.

My reasoning and evidence

6. W2 Trace separate calls

Using the correct definition, store `a = remaining(8, 3)`, `b = remaining(8, 8)`, `c = remaining(8, 3)`. State `a`, `b` and `c`. In a fresh script, explain why `print(spaces)` outside the function fails.

My reasoning and evidence

7. W3 Repair the returned value

Use the faulty printing version with `answer = remaining(11, 2)`. State the displayed number and the value stored in `answer`. Explain why `print(answer + 1)` fails, then repair the function and predict the repaired output.

My reasoning and evidence

8. W4 Build a reusable calculation independently

Each tray holds 5 counters. Define `stock(trays, loose)` to return the total using a local variable `total`. Assume nonnegative integer arguments. Store results for `(2, 3)`, `(0, 3)` and `(2, 0)`. Print the sum of the first two stored results. Predict that output and all three stored values. Call `(2, 3)` again to check fresh state.

My reasoning and evidence

9. W5 Locate a statement that never runs

A function body has spaces = capacity - used, then return spaces, then print("done"), all at the same indentation. What does answer = remaining(4, 4) store and display? Explain whether the final print can run.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled return values and local state evidence sheet to Python readiness portfolio. Show how you can use a returned value in a later calculation. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Return values and local state

Return values survive their calls. A function returns a local calculation that the caller stores.

Python 3: `def extra(a,b): result=a+b; return result. Then first=extra(2,5); second=extra(4,1); print(first,second).`

Return values and local state

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Trace first call

Its local result is 7; return assigns 7 to first.



2. Trace second call

Its new local result is 5; return assigns 5 to second.



3. Read caller values

The final print displays 7 5; the second call does not overwrite first.

CHANGE ONE THING

Add `print(result)` after the calls in a fresh script with no global result.

Explain the changed result using the matching lesson.

Trace the two returned values and explain why first stays seven. What happens if the caller tries to print result directly?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

13. Project portfolio milestone

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, predictions, actual outputs, one repaired error and an explanation of a function on changed inputs.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Return values survive their calls. A function returns a local calculation that the caller stores.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The final print displays 7 5; the second call does not overwrite first.

1. Read caller values
2. Trace first call
3. Trace second call

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Read caller values The final print displays 7 5; the second call does not overwrite first. Returning passes a value back to the caller. Local names belong to a function call; storing the returned value creates a separate caller binding.

Practice 2: explain the deciding evidence

Changed-case practice: Add print(result) after the calls in a fresh script with no global result. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. NameError occurs because that local name is not defined in the caller.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: `NameError` occurs because that local name is not defined in the caller. A returned value and the function's internal name are different things.

Repair a missing building block

Review Functions and arguments (python-bridge/8). Goal: Distinguish a function definition from a function call. Short practice: Match every argument to a parameter for the two calls. Why do outputs differ, and what prints when both calls are removed?. Return to this week and explain what you changed.

Review Loops and trace tables (python-bridge/6). Goal: Trace variable values before and after each loop iteration. Short practice: List the four visited values and their running totals. Explain why 5 is absent and what prints for `range(1,1)`.. Return to this week and explain what you changed.

Review Expressions and operator order (python-bridge/3). Goal: Evaluate numeric expressions using Python's grouping rules. Short practice: Trace the operation order in both expressions. Identify which picture step changes with parentheses and predict both results when the multiplier becomes four.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
